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Provenance and Local Causal Models

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- Local Causal Models
- Building from Multiple LCMs
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- Overlapping Claims
- Edge Metadata
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Introduction

A key feature of Csql is **provenance tracking**: every causal claim is linked back to its source document and local causal model (LCM). This enables:

- Auditing which papers support a claim
- Weighting evidence by document quality
- Detecting when claims come from a single vs. multiple sources
- Understanding evidence aggregation

This vignette covers the `LocalCausalModel` type, the `atlas_edge_support` table, and provenance queries.

Local Causal Models

A *Local Causal Model* (LCM) is a set of causal triples extracted from a single document or hypothesis. Instead of adding triples one by one, you can package them as an LCM:

```
using CSQL
```

```
# LCM from a transmission dynamics paper
```

```
lcm1 = LocalCausalModel("lcm_trans_001", "smith2023",  
  [  
    CausalTriple("Pathogen virulence", "increases", "Disease severity"),  
    CausalTriple("Pathogen transmissibility", "increases", "Basic reproduction number"),  
    CausalTriple("Basic reproduction number", "increases", "Epidemic potential"),  
    CausalTriple("Disease severity", "increases", "Case fatality rate"),  
  ];  
focus="transmission dynamics",  
score=0.92,  
metadata=Dict{String,Any}("stage" => "validated", "source_model" => "GPT-4")  
)  
println("LCM: $(lcm1.lcm_id) from $(lcm1.doc_id)")  
println("  Focus: $(lcm1.focus)")  
println("  Triples: $(length(lcm1.triples))")  
println("  Score: $(lcm1.score)")
```

```
LCM: lcm_trans_001 from smith2023  
Focus: transmission dynamics  
Triples: 4  
Score: 0.92
```

Building from Multiple LCMs

Let's build an atlas from multiple LCMs representing different papers:

```
# LCM from an immunity paper
```

```
lcm2 = LocalCausalModel("lcm_immun_001", "chen2023",  
  [  
    CausalTriple("Vaccination", "increases", "Antibody levels"),  
    CausalTriple("Antibody levels", "reduces", "Susceptibility"),  
    CausalTriple("Natural infection", "increases", "Antibody levels"),  
    CausalTriple("Susceptibility", "increases", "Basic reproduction number"),  
  ];  
focus="immunity",  
score=0.88,  
metadata=Dict{String,Any}("stage" => "original", "source_model" => "Claude")  
)
```

```
# LCM from an intervention paper - overlaps with lcm1
```

```
lcm3 = LocalCausalModel("lcm_interv_001", "patel2024",  
  [  
    CausalTriple("Healthcare capacity", "reduces", "Case fatality rate"),  
    CausalTriple("Disease severity", "increases", "Case fatality rate"),  
    CausalTriple("Quarantine", "reduces", "Basic reproduction number"),  
  ]
```

```

];
focus="interventions",
score=0.85,
metadata=Dict{String,Any}("stage" => "original", "source_model" => "GPT-4")
)

# Build atlas from all three LCMs
builder = AtlasBuilder()
add_lcm!(builder, lcm1)
add_lcm!(builder, lcm2)
add_lcm!(builder, lcm3)

csql = connect_csql()
build!(builder, csql.db)

stats = statistics(csql)
println("Atlas: $(stats[:n_nodes]) nodes, $(stats[:n_edges]) edges, $(stats[:n_support]) support records")
Atlas: 12 nodes, 10 edges, 11 support records

```

Querying Provenance

The atlas_edge_support table tracks which documents and LCMs support each edge.

All Support Records

```

support = custom_query(csql, """
    SELECT es.doc_id, es.lcm_instance_id, es.score,
           n1.label_canon AS src, n2.label_canon AS dst
    FROM atlas_edge_support es
    JOIN atlas_edges e ON es.edge_id = e.edge_id
    JOIN atlas_nodes n1 ON e.src_id = n1.node_id
    JOIN atlas_nodes n2 ON e.dst_id = n2.node_id
    ORDER BY es.doc_id, es.lcm_instance_id
""")
println("Total support records: $(length(support))")
for s in support
    println("    [$(s.doc_id)/$(s.lcm_instance_id)] $(s.src) → $(s.dst)    score=$(round(s.score, 2))")
end

Total support records: 11
[chen2023/lcm_immun_001] vaccination → antibody levels    score=0.88
[chen2023/lcm_immun_001] antibody levels → susceptibility score=0.88
[chen2023/lcm_immun_001] natural infection → antibody levels    score=0.88
[chen2023/lcm_immun_001] susceptibility → basic reproduction number    score=0.88
[patel2024/lcm_interv_001] healthcare capacity → case fatality rate    score=0.85

```

```
[patel2024/lcm_interv_001] disease severity → case fatality rate score=0.85
[patel2024/lcm_interv_001] quarantine → basic reproduction number score=0.85
[smith2023/lcm_trans_001] pathogen virulence → disease severity score=0.92
[smith2023/lcm_trans_001] pathogen transmissibility → basic reproduction number score=0.92
[smith2023/lcm_trans_001] basic reproduction number → epidemic potential score=0.92
[smith2023/lcm_trans_001] disease severity → case fatality rate score=0.92
```

Edges Supported by Multiple Documents

Claims supported by multiple independent sources are more credible:

```
multi_doc = custom_query(csql, """
    SELECT e.rel_type, n1.label_canon AS src, n2.label_canon AS dst,
           e.support_docs, e.support_lcms, e.score_sum
    FROM atlas_edges e
    JOIN atlas_nodes n1 ON e.src_id = n1.node_id
    JOIN atlas_nodes n2 ON e.dst_id = n2.node_id
    WHERE e.support_docs > 1
    ORDER BY e.support_docs DESC
""")
println("Edges with multi-document support:")
for m in multi_doc
    println("  $(m.src) -[$(m.rel_type)]→ $(m.dst)")
    println("    docs=$(m.support_docs), lcms=$(m.support_lcms), total_score=$(round(m.score, 2))")
end
if isempty(multi_doc)
    println("  (none - each edge comes from a single document in this example)")
end

Edges with multi-document support:
disease severity -[INCREASES]→ case fatality rate
docs=2, lcms=2, total_score=1.77
```

Provenance by Document

Which claims does each document contribute?

```
by_doc = custom_query(csql, """
    SELECT es.doc_id,
           COUNT(DISTINCT es.edge_id) AS n_edges,
           COUNT(*) AS n_records,
           SUM(es.score) AS total_score
    FROM atlas_edge_support es
    GROUP BY es.doc_id
    ORDER BY total_score DESC
""")
println("Provenance by document:")
for d in by_doc
```

```

        println("  $(d.doc_id): $(d.n_edges) edges, $(d.n_records) support records, score=$(rou
end

```

Provenance by document:

```

smith2023: 4 edges, 4 support records, score=3.68
chen2023: 4 edges, 4 support records, score=3.52
patel2024: 3 edges, 3 support records, score=2.55

```

Overlapping Claims

When multiple LCMs assert the same causal relationship, CSQL.jl aggregates them. Let's find these overlaps:

```

overlaps = custom_query(csql, """
    SELECT e.rel_type, n1.label_canon AS src, n2.label_canon AS dst,
           e.support_lcms, e.score_sum, e.score_mean, e.score_max
    FROM atlas_edges e
    JOIN atlas_nodes n1 ON e.src_id = n1.node_id
    JOIN atlas_nodes n2 ON e.dst_id = n2.node_id
    WHERE e.support_lcms > 1
    ORDER BY e.support_lcms DESC
""")
println("Aggregated edges (multiple LCM support):")
for o in overlaps
    println("  $(o.src) -[$(o.rel_type)]→ $(o.dst)")
    println("    lcms=$(o.support_lcms), sum=$(round(o.score_sum; digits=2)), mean=$(round(o
end

```

```

Aggregated edges (multiple LCM support):
disease severity -[INCREASES]→ case fatality rate
lcms=2, sum=1.77, mean=0.88, max=0.92

```

Edge Metadata

The `atlas_edges` table stores rich metadata about each causal claim:

```

edges = custom_query(csql, """
    SELECT e.rel_type, n1.label_canon AS src, n2.label_canon AS dst,
           e.stage, e.confidence, e.grounded, e.source_model
    FROM atlas_edges e
    JOIN atlas_nodes n1 ON e.src_id = n1.node_id
    JOIN atlas_nodes n2 ON e.dst_id = n2.node_id
    ORDER BY e.score_sum DESC
    LIMIT 5
""")
println("Top 5 edges with metadata:")
for e in edges
    println("  $(e.src) → $(e.dst)")

```

```

println("    stage=$(e.stage), confidence=$(e.confidence), grounded=$(e.grounded), model=$(e.model)"
end

```

Top 5 edges with metadata:

```

disease severity → case fatality rate
  stage=validated, confidence=1.0, grounded=not_evaluated, model=GPT-4
pathogen transmissibility → basic reproduction number
  stage=validated, confidence=1.0, grounded=not_evaluated, model=GPT-4
pathogen virulence → disease severity
  stage=validated, confidence=1.0, grounded=not_evaluated, model=GPT-4
basic reproduction number → epidemic potential
  stage=validated, confidence=1.0, grounded=not_evaluated, model=GPT-4
susceptibility → basic reproduction number
  stage=original, confidence=1.0, grounded=not_evaluated, model=Claude

```

Summary

This vignette demonstrated:

- **LocalCausalModel** — packaging triples from a single source
- **add_lcm!** — batch ingestion of causal models
- **atlas_edge_support** — provenance table linking edges to documents
- **Multi-document evidence** — aggregation across independent sources
- **Edge metadata** — stage, confidence, grounding status, source model

Next: Atlas Merging covers combining causal databases from different research domains.